

THOMPSON, MB, Canada

WMO: 710790

Lat: 55.800N

Lon: 97.860W

Elev: 735

StdP: 14.31

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.3	-32.9	-44.5	0.4	-37.1	-40.9	0.5	-32.6	20.0	-4.2	18.2	-5.4	3.2	300	0.738

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.0	82.2	64.3	79.2	62.9	76.3	61.4	67.3	77.5	65.2	74.8	63.3	72.6	10.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.8	91.0	71.7	61.7	84.6	69.3	59.5	78.0	67.4	32.3	77.4	30.6	74.8	29.1	72.8	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.7	17.7	15.7													
			DB	-43.8	89.1	4.6	3.4	-47.1	91.5	-49.8	93.4	-52.4	95.3	-55.7	97.8
			WB	-43.8	70.9	4.4	2.3	-47.0	72.6	-49.6	74.0	-52.1	75.3	-55.3	77.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.8	-9.1	-4.2	9.4	27.9	42.8	55.8	61.6	58.4	47.6	33.2	12.2	-3.8
	DBStd	27.36	13.87	13.11	14.58	12.97	9.56	7.82	5.97	6.58	8.26	8.43	13.24	14.84
	HDD50	9043	1832	1516	1258	667	265	31	1	7	142	522	1134	1668
	HDD65	13635	2297	1936	1723	1114	689	289	139	224	523	984	1584	2133
	CDD50	952	0	0	0	2	42	205	361	268	71	3	0	0
	CDD65	70	0	0	0	0	1	13	34	20	2	0	0	0
	CDH74	1126	0	0	0	0	39	260	500	285	42	0	0	0
	CDH80	203	0	0	0	0	7	46	93	53	4	0	0	0
Wind	WSAvg	7.2	6.8	7.1	7.5	8.0	8.4	7.6	6.9	7.1	7.2	7.3	6.6	6.2
Precipitation	PrecAvg	20.9	0.9	0.7	0.9	1.0	1.7	2.5	3.6	3.5	2.5	1.7	1.2	0.7
	PrecMax	34.4	2.0	1.8	2.1	3.2	4.0	5.7	8.3	10.1	4.7	4.4	4.0	1.8
	PrecMin	15.0	0.2	0.1	0.1	0.0	0.3	0.3	1.3	0.9	0.7	0.7	0.4	0.2
	PrecStd	3.9	0.5	0.3	0.5	0.8	0.9	1.2	1.7	2.1	1.2	0.8	0.8	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.3	34.0	47.9	67.3	78.3	85.1	87.1	85.4	79.1	62.8	41.7	30.7
		MCWB	29.2	32.0	39.3	50.0	57.9	64.6	67.3	66.9	64.0	52.1	37.4	29.4
	2%	DB	23.3	28.9	42.4	58.7	72.5	80.2	82.4	80.8	72.9	55.6	36.2	26.0
		MCWB	22.1	27.1	35.7	45.7	54.8	61.4	64.8	64.8	59.9	48.1	33.8	24.9
	5%	DB	18.3	22.6	36.6	52.3	67.2	76.8	79.4	77.1	67.3	50.0	32.9	22.1
		MCWB	17.5	21.0	32.0	41.3	51.8	59.9	63.3	63.4	57.2	44.8	31.5	21.2
	10%	DB	12.8	17.3	31.7	47.2	61.6	73.1	76.5	72.8	62.3	45.5	30.2	18.6
		MCWB	12.0	16.0	28.9	38.4	49.3	58.4	62.2	61.1	54.5	42.2	28.8	18.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.0	32.2	39.6	51.4	59.9	69.2	70.1	69.2	65.2	53.4	37.5	29.5
		MCDB	29.9	34.2	46.2	65.0	74.4	78.3	81.3	80.2	75.1	59.0	40.3	30.6
	2%	WB	22.3	27.2	36.1	46.4	56.2	65.0	67.4	66.8	61.3	48.9	34.1	25.0
		MCDB	23.3	29.0	41.7	57.8	69.4	74.6	78.0	77.0	71.2	54.8	35.8	25.9
	5%	WB	17.6	21.0	32.5	42.4	53.1	62.0	65.4	64.8	58.2	45.4	31.7	21.3
		MCDB	18.3	22.4	36.1	51.1	65.3	73.6	75.2	73.7	65.5	49.4	32.9	22.2
	10%	WB	12.1	15.9	28.7	38.8	50.0	59.4	63.5	62.7	55.2	42.3	28.8	17.9
		MCDB	12.9	17.3	31.9	46.4	60.9	70.2	73.0	71.0	62.4	45.5	30.1	18.7
Mean Daily Temperature Range	5% DB	MDBR	17.2	22.1	25.2	24.0	23.3	24.0	23.0	21.7	19.9	13.6	14.9	16.2
		MCDBR	17.5	21.9	25.5	29.9	33.8	30.9	28.9	28.4	28.4	23.0	15.1	15.8
		MCWBR	17.1	20.2	19.5	18.8	18.9	15.3	14.3	15.0	16.8	15.7	13.1	15.4
	5% WB	MCDBR	17.3	20.6	22.9	27.8	30.0	26.2	24.3	24.4	25.0	20.7	14.5	15.6
		MCWBR	16.9	19.1	17.7	17.9	17.4	13.9	13.3	14.0	15.9	14.7	12.7	15.2
Clear-Sky Solar Irradiance	taub		0.242	0.241	0.276	0.318	0.334	0.363	0.402	0.359	0.321	0.287	0.250	0.219
	taud		2.193	2.314	2.274	2.344	2.453	2.390	2.329	2.467	2.525	2.572	2.352	2.194
	Ebn at Noon		218	272	285	287	288	279	265	271	269	255	219	200
	Edh at Noon		17	23	31	34	32	35	37	30	24	18	15	13
All-Sky Solar Radiation	RadAvg		261	610	1110	1625	1769	1826	1701	1370	924	478	227	165
	RadStd		24	43	88	128	153	162	119	116	108	66	22	24

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page